

Nonlinear Methods

Faculty of Information Technology, Monash University, Australia

FIT5149 week 3

- 1 Assignment 1
- 2 Simple Nonlinear Extension of Linear Models
- 3 Regression Splines
- 4 Smooth Splines
- 5 Summary

Sanctioned Loan Amount Prediction

As a data scientist at **Yarra Credit Group**, an Australian consumer lender, you will build a regression model that predicts `sanctioned_amount_aud`, the amount sanctioned on each new loan application.

Scenario. The amount an applicant requests and the amount the lender commits are rarely the same: the lender weighs capacity to service the debt, the security offered, credit history, and its own risk appetite. Your model automates part of this judgement from a historical extract assembled from several internal systems and **never cleaned**; establishing what the data actually contains carries a substantial share of the marks.

- **Due:** 11:55 PM, Monday 14 September 2026 (Kaggle closes at the same time)
- **Groups:** Two students, self-managed on Moodle, finalised by Sat 5 September
- **Weight:** 25% of the unit; the report carries 70 of the 100 marks
- **Language:** Python or R; late penalty 5% per day, up to 7 days

Your Tasks

Prediction task (regression): predict the sanctioned amount in AUD.

- Compare **at least three different types** of model; feature sets or hyperparameter settings of one algorithm count as a single type.
- Apply appropriate preprocessing, feature engineering and feature selection.
- Design and justify a validation strategy and use **it, not the public leaderboard**, to choose the final model; retrain it on all labelled data for the test predictions.

Analysis task:

- Establish the condition of the data: what you found, what you did about it, and what effect it had on your results.
- Identify the features that drive the sanctioned amount, with statistical evidence beyond a single feature-importance plot.
- Discuss the limitations of your final model and where it performs poorly.

Metric: RMSE in AUD, the typical dollar error per application; lower is better.

The Data



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Training: 19,322 labelled applications, 29 columns (including target)

Test: 10,000 unlabelled applications for Kaggle submission

Variable types: binary, nominal, ordinal, numeric and date

Applicant & Income	Credit & Security
Age, gender, dependants	Credit rating, prior defaults
Annual & monthly income	Credit enquiries, card status
Occupation, employment sector	Property value, age, category
Existing repayments, expense flags	Property area, risk review flag
Requested amount, residence area	Branch, channel, application date

Kaggle Competition (30/100 marks)

FIT5149 S2 2026 Ass1 - Sanctioned Loan Amount Prediction

Predict the sanctioned loan amount (in AUD) for consumer loan applications.

[Overview](#) [Data](#) [Code](#) [Models](#) [Discussion](#) [Leaderboard](#) [Rules](#)

- **Two leaderboards.**
 - ▶ Public score on 2,000 test rows;
 - ▶ Private score on the remaining 8,000, hidden until close. **Marks use the private score only.**
- **The public leaderboard is not a validation set.**
- **Limits.** One Kaggle team per group, named in the report; 3 submissions per day; nominate up to two final entries before the close.



Learning Outcomes

After working your way through this module, you should be able to:

- Understand and assess different nonlinear models, such as polynomial regression and splines.
- Explain the limits and constraints of nonlinear methods

aligned with unit learning outcomes

- Evaluate the limitations, appropriateness and benefits of data analytics methods for given tasks;
- Design solutions to real world problems with data analytics techniques;



Motivations

- Any real-world examples of nonlinear relationship?
- What nonlinear functions you know
- Which are more fundamental (e.g., used to approximate general functions)

Motivations

- Any real-world examples of nonlinear relationship?
 - ▶ Weight loss amount of exercises
 - ▶ Height of sea tides time
 - ▶ Object classes pixels in images
 - ▶ Hospital readmission rate electronic health records
- What nonlinear functions you know
 - ▶ Quadratic
 - ▶ Cubic
 - ▶ Exponential
 - ▶ Cosine
- Which are more fundamental (e.g., used to approximate general functions)

Polynomial Regression: the basics

- A polynomial function

$$\begin{aligned}
 y_i &= \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \cdots + \beta_d x_i^d + \epsilon_i \\
 &= \sum_{j=0}^d \beta_j x_i^j
 \end{aligned}$$

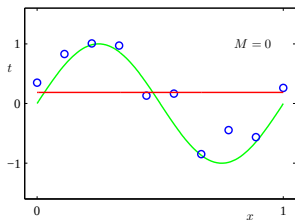
where d is the order (or degree) of the polynomial.

- ▶ Constant polynomial: β_0
- ▶ linear polynomial: $\beta_0 + \beta_1 x_i$
- ▶ quadratics: $\beta_0 + \beta_1 x_i + \beta_2 x_i^2$
- ▶ cubics: $\beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3$
- Curve fitting: minimise the following error function

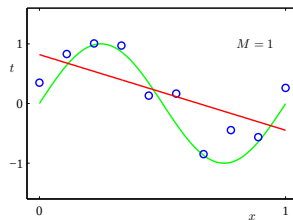
$$E(\boldsymbol{\beta}) = \frac{1}{2} \sum_{i=1}^N (\hat{y}_i - y_i)^2 = \frac{1}{2} \sum_{i=1}^N \left(\sum_{j=0}^d \beta_j x_i^j - y_i \right)^2$$

- Used to approximate continuous real-valued functions in $[a, b]$ (Weierstrass approximation theorem).

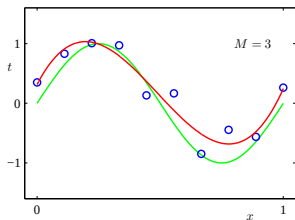
Polynomial Regression: fit data generated with $\sin()$



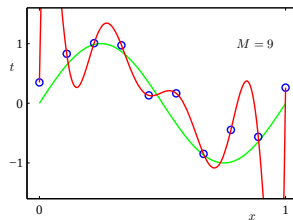
(a)



(b)



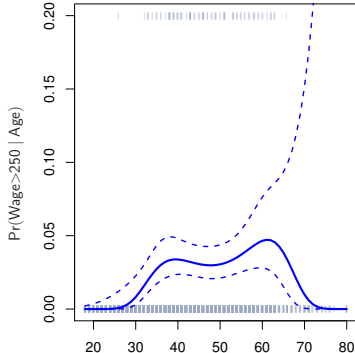
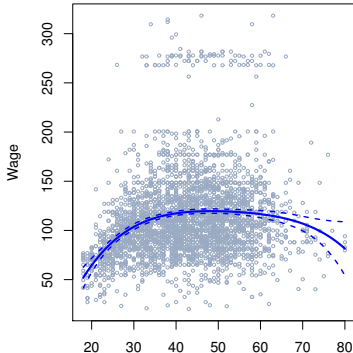
(c)



(d)

Overfitting VS. underfitting

Polynomial Regression: Predict Wage with Age

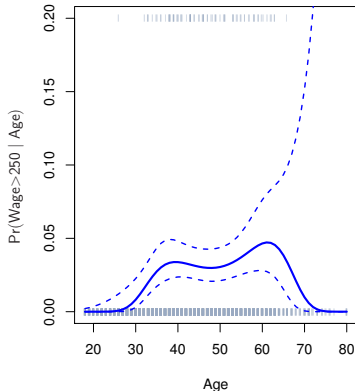
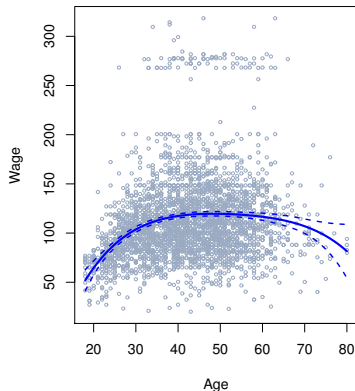


- Task: predict wage based on income and demographic information
- Polynomial function :

$$\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0 + \hat{\beta}_2 x_0^2 + \hat{\beta}_3 x_0^3 + \hat{\beta}_4 x_0^4$$

- ▶ $\hat{f}(x_0)$ is a linear function of the $\hat{\beta}_i$
- ▶ The dashed curves indicate an estimated 95% confidence interval

Polynomial Regression: Predict Wage with Age



- Logistic regression

$$\Pr(y_i > 250 \mid x_i) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x_0 + \hat{\beta}_2 x_0^2 + \hat{\beta}_3 x_0^3 + \hat{\beta}_4 x_0^4}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 x_0 + \hat{\beta}_2 x_0^2 + \hat{\beta}_3 x_0^3 + \hat{\beta}_4 x_0^4}}$$

- The confidence interval is fairly wide due to that there are only 79 high earners.

Step functions

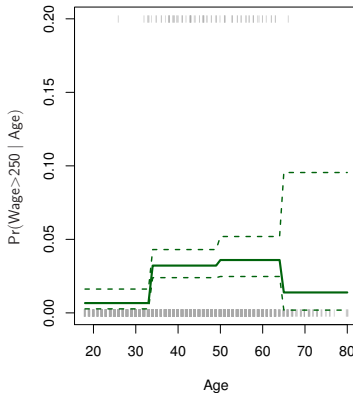
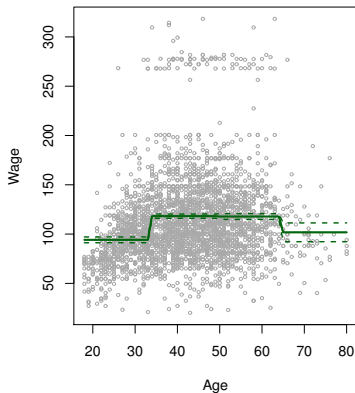
- Another way of creating transformations of a variable — cut the variable into distinct regions and *avoid impose a global structure*.
 - Create cut-points $c_1, c_2, c_3, \dots, c_K$, and $K + 1$ variables

$$\begin{aligned}
 C_0(X) &= I(X < c_1), \\
 C_1(X) &= I(c_1 \leq X < c_2), \\
 C_2(X) &= I(c_2 \leq X < c_3), \\
 &\vdots \\
 C_{K-1}(X) &= I(c_{K-1} \leq X < c_K), \\
 C_K(X) &= I(c_K \leq X),
 \end{aligned}$$

- Fit a linear model using those variables

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \beta_3 C_3(x_i) + \dots + \beta_K C_K(x_i) + \epsilon_i$$

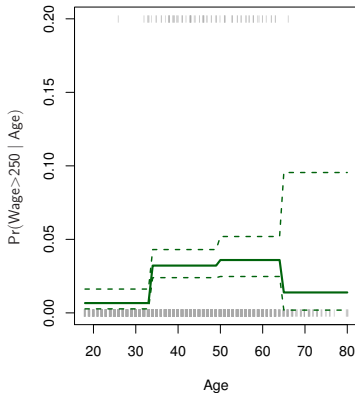
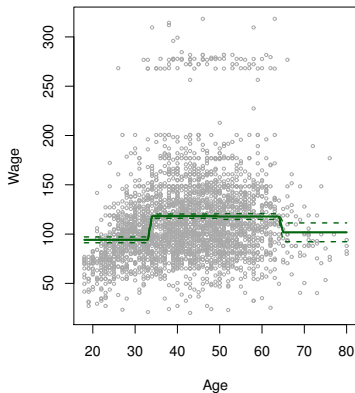
Step functions



- Easy to create a series of dummy variables representing each group

$$C_1(X) = I(X < 35) \quad C_2(X) = I(35 \leq X < 50) \quad C_3(X) = I(50 \leq X < 65), \dots$$

Step functions



- Choice of cut-points or knots can be problematic. For creating nonlinearities, smoother alternatives such as splines are available.



Basis Functions

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \beta_3 b_3(x_i) + \cdots + \beta_K b_K(x_i) + \epsilon_i$$

- $b_k(X)$: a function or transformation that can be applied to variable X .
- What is $b_k(x_i)$ for the polynomial regression?
- What is $b_k(x_i)$ for the piecewise constant functions?

Basis Functions

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \beta_3 b_3(x_i) + \cdots + \beta_K b_K(x_i) + \epsilon_i$$

- $b_k(X)$: a function or transformation that can be applied to variable X .
- What is $b_k(x_i)$ for the polynomial regression?

$$b_k(x_i) = x_i^k$$

- What is $b_k(x_i)$ for the piecewise constant functions?

$$b_k(x_i) = I(c_k \leq x_i^k \leq c_{k+1})$$



Activity 1: Polynomial Regression

Apply polynomial regression to the single feature model with *GarageCars* and identify which degree achieves the lowest CV errors.

Outline



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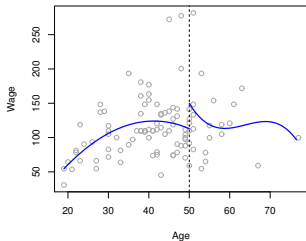
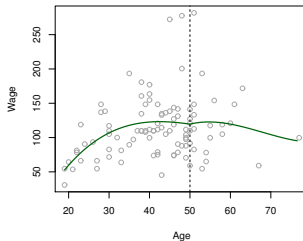
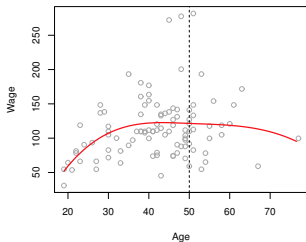
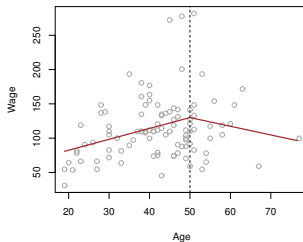
Piecewise Polynomials

- Instead of a single polynomial in X over its whole domain, we can rather use different polynomials in regions defined by knots (c). For example, a piecewise cubic polynomials over different region of X :

$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta_{21}x_i^2 + \beta_{31}x_i^3 + \epsilon_i & \text{if } x_i < c; \\ \beta_{02} + \beta_{12}x_i + \beta_{22}x_i^2 + \beta_{32}x_i^3 + \epsilon_i & \text{if } x_i \geq c; \end{cases}$$

- Knots: the points where the coefficients change
- Using more knots leads to a more flexible piecewise polynomial.
 - ▶ K different knots $\rightarrow K + 1$ different cubic polynomials
- Key assumptions:
 - ▶ The true functions are continuous
 - ▶ The true functions are smooth

Piecewise Polynomials: continuity and smooth

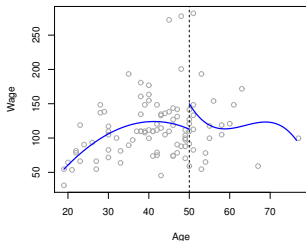
Piecewise Cubic**Continuous Piecewise Cubic****Cubic Spline****Linear Spline**

Piecewise Polynomials: continuity and smooth

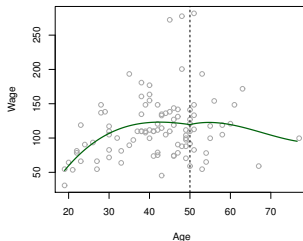
- Continuity
- Continuity of the first derivative (change rates).
what is the geometry of the first derivative?
 - ▶ if $\frac{df}{dx}(p) > 0$, then $f(x)$ is an increasing function at $x = p$
 - ▶ if $\frac{df}{dx}(p) < 0$, then $f(x)$ is an decreasing function at $x = p$
 - ▶ if $\frac{df}{dx}(p) = 0$, then $x = p$ is called a critical point of $f(x)$
- Continuity of the second derivative (change rates of changes).
 - ▶ if $\frac{d^2f}{dx^2}(p) > 0$ at $x = p$, then $f(x)$ is convex at $x = p$
 - ▶ if $\frac{d^2f}{dx^2}(p) < 0$ at $x = p$, then $f(x)$ is concave at $x = p$
 - ▶ if $\frac{d^2f}{dx^2}(p) = 0$ at $x = p$, p is an inflection point, local maximum, or local minimum

Piecewise Polynomials: continuity and smooth

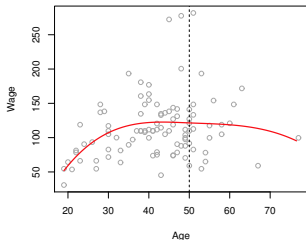
Piecewise Cubic



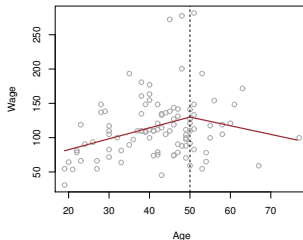
Continuous Piecewise Cubic



Cubic Spline



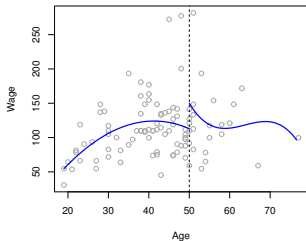
Linear Spline



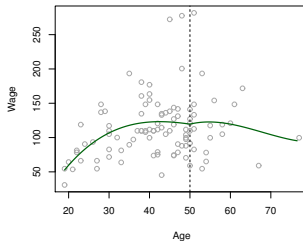
DF for fitting the model: top_left $\rightarrow 8$; bottom_left $\rightarrow 5$

Piecewise Polynomials: continuity and smooth

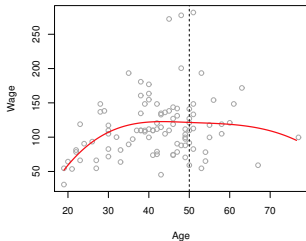
Piecewise Cubic



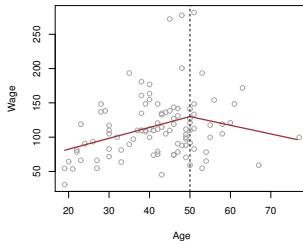
Continuous Piecewise Cubic



Cubic Spline



Linear Spline



DF for fitting the model with K knots: $4 + K$

Linear Splines

- A linear spline with knots at ξ_k , $k = 1, \dots, K$ is a piecewise linear polynomial continuous at each knot

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \dots + \beta_{K+1} b_{K+1}(x_i) + \epsilon_i$$

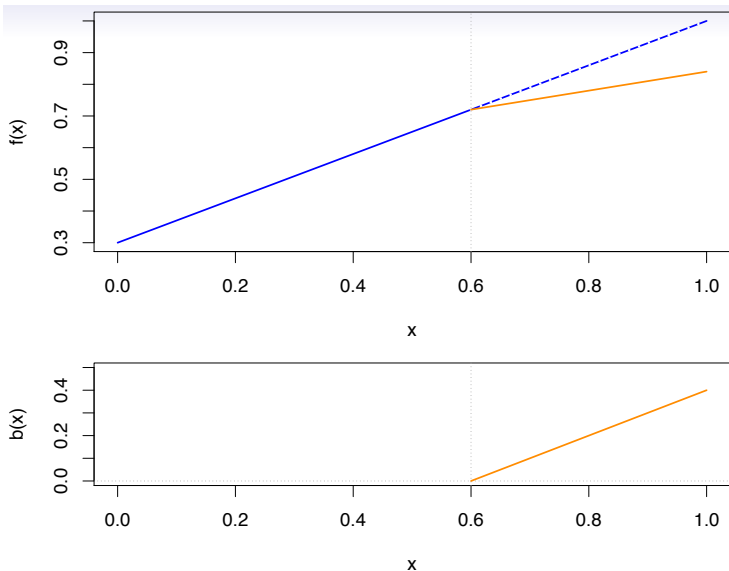
where the b_k are basis functions

$$\begin{aligned} b_1(x_i) &= x_i \\ b_{k+1}(x_i) &= (x_i - \xi_k)_+, \quad k = 1, \dots, K \end{aligned}$$

and

$$(x_i - \xi_k)_+ = \begin{cases} x_i - \xi_k & \text{if } x_i > \xi_k \\ 0 & \text{otherwise} \end{cases}$$

Linear Splines



Cubic Splines

- A cubic spline with knots at ξ_k , $k = 1, \dots, K$ is a piecewise cubic polynomial with continuous derivatives up to order 2 at each knot.

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \dots + \beta_{K+3} b_{K+3}(x_i) + \epsilon_i$$

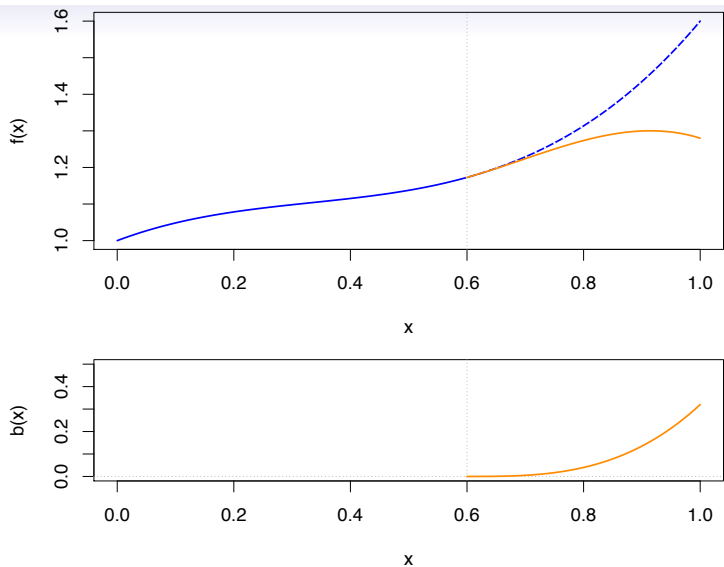
where

$$\begin{aligned} b_1(x_i) &= x_i \\ b_2(x_i) &= x_i^2 \\ b_3(x_i) &= x_i^3 \\ b_{k+1}(x_i) &= (x_i - \xi_k)_+^3, \quad k = 1, \dots, K \end{aligned}$$

and

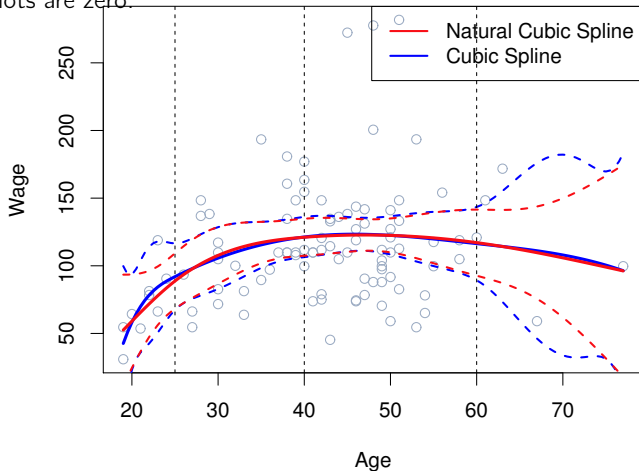
$$(x_i - \xi_k)_+^3 = \begin{cases} (x_i - \xi_k)^3 & \text{if } x_i > \xi_k \\ 0 & \text{otherwise} \end{cases}$$

Cubic Splines



Natural Cubic Splines

- A natural cubic spline extrapolates linearly beyond the boundary knots.
 - ▶ Add extra constraints to the end, i.e., the second derivatives at the two outer knots are zero.



#Knots and Knot placement

- One strategy is to decide K , and then place them at appropriate quantiles of the observed X

Natural Cubic Spline

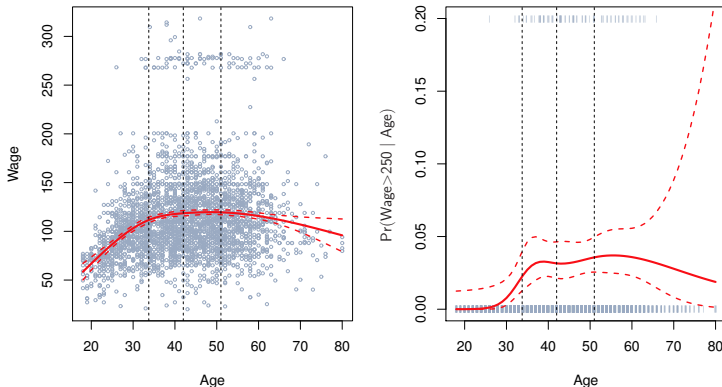
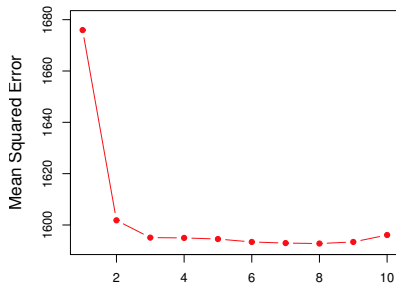


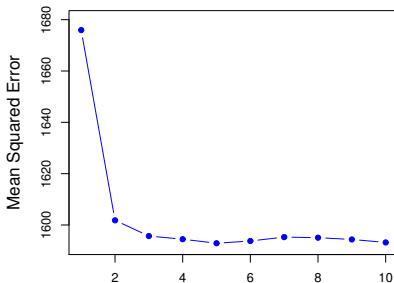
Figure: 3 knots are chosen automatically as the 25th, 50th and 75th percentiles. (3 knots correspond to 4 DF for fitting the natural cubic splines.)

#Knots and Knot placement

- Knots determine where the spline is allowed to change its shape and therefore control the flexibility of the fitted curve.
- A common strategy is to place knots at selected quantiles of X , so that each region contains roughly the same number of observations.
- The number of knots can be selected using cross-validation.



Degrees of Freedom of Natural Spline



Degrees of Freedom of Cubic Spline

Figure: 10-fold cross-validation for selecting the degrees of freedom.

#Knots and Knot placement

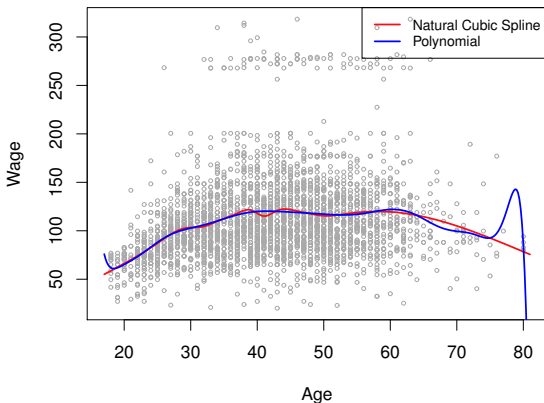


Figure: A degree-15 Polynomial vs a natural cubic spline with 15 degrees of freedom.

- Polynomial uses a high degree to produce flexible fit.
- Splines introduce flexibility by increasing K but keeping the degree fixed.
- Splines allow us to place more knots, and hence flexibility, over regions where the function f seems to be changing rapidly.



Activity 2: Splines

Apply B-Splines with appropriate knots and polynomial degree to the predictor *GarageCars*. Identify which knots and degree achieve the lowest RMSE via CV, together with the remaining features.

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Smooth Splines: overview

- Regression splines:

- ▶ specify a set of knots,
- ▶ produce a sequence of basis functions, and
- ▶ use least squares in fitting

$$RSS = \sum_{i=1}^n (y_i - g(x_i))^2$$

- Problem: If we don't have any constraints on $g(x_i)$, we can always make RSS zero simply by choosing g such that it interpolates all of y_i .
- What we want
 - ▶ Small RSS, and
 - ▶ Smooth fitted function

Smooth Splines: target function

- Smoothing spline

$$\underset{g \in \mathcal{S}}{\text{minimize}} \sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$$

- ▶ The first term is RSS, and tries to make $g(x)$ match the data at each x_i
- ▶ The second term is a roughness penalty and controls how wiggly $g(x)$ is.
- ▶ Tuning parameter $\lambda \geq 0$:
 - The smaller λ , the more wiggly the function, eventually interpolating y_i when $\lambda = 0$.
 - As $\lambda \rightarrow \infty$, the function $g(x)$ becomes linear.
- ▶ Note on derivatives
 - $g'(t)$: measures the slope of a function at t
 - $g''(t)$: measures the amount by which the slope is changing. If it is large in absolute value if $g(t)$ is very wiggly near t ; it is close to zero otherwise.
- ▶ $\int g''(t)^2 dt$: a measure of the total change in the function $g'(t)$.
 - If $g(t)$ is smooth, $\int g''(t)^2 dt$ will take on a small value.
 - If $g(t)$ is jumpy and variable, $\int g''(t)^2 dt$ will take on a large value.

Smooth Splines: target function

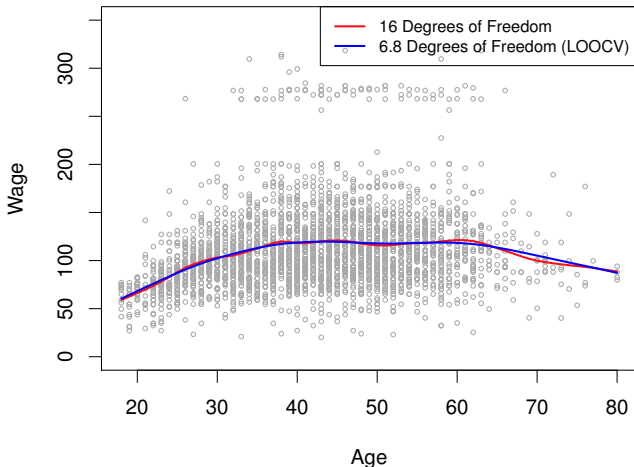
- Smoothing spline

$$\underset{g \in \mathcal{S}}{\text{minimize}} \sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$$

- ▶ The first term is RSS, and tries to make $g(x)$ match the data at each x_i
- ▶ The second term is a roughness penalty and controls how wiggly $g(x)$ is.
- ▶ Tuning parameter $\lambda \geq 0$:
 - The smaller λ , the more wiggly the function, eventually interpolating y_i when $\lambda = 0$.
 - As $\lambda \rightarrow \infty$, the function $g(x)$ becomes linear.
- ▶ A natural cubic spline with a knot at every unique value of x_i
- ▶ Some details
 - Smoothing splines avoid the knot-selection issue
 - Cross-validate a single parameter λ

Smooth Splines

Smoothing Spline



- Choosing the Smoothing Parameter using CV (Optional).

Summary

- Simple extensions of linear models, e.g, polynomial regression, step functions, etc.
- Splines: Regression splines & Smooth splines
- Reading materials:
 - ▶ "Moving Beyond Linearity", Chapter 7 of "Introduction to Statistical Learning with Applications in Python/R".
- Post-class activities:
 - ▶ Encouraged to attempt the Lab (Sec. 7.8) of the textbook and the exercises (Sec. 7.9).
- Acknowledgement:
 - ▶ Figures in this presentation were taken from "An Introduction to Statistical Learning, with applications in Python/R" (Springer, 2023) with permission from the authors: G. James, D. Witten, T. Hastie and R. Tibshirani
 - ▶ Some of the slides are reproduced based on the slides from T. Hastie and R. Tibshirani
 - ▶ Figures on the slide 7 are from Christopher Bishop's website.